

Physics layer Φ

(frozen after pretraining)

self-supervised dynamics · behavior proposals



Semantic layer Ω

(revisable)

taxonomy evolves $\mathcal{Y} \rightarrow \mathcal{Y}'$: only Ω retrains



budget retention

NTU60

90.7%

NTU120

88.9%

canine



28.9%

retention of own full-budget top-1 at 10% labels;

canine: 9.8 vs 11.1% chance at 13% budget